Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/21/2006
PI Name:	Simons, Donald Ph.D.		
Project Title:	Impaired Activation and Signal Transduction in Mouse Splenocytes Cultured in Modeled Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dms42@drexel.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-895-1932
Organization Name:	Drexel University		
PI Address 1:	School of Biomed, Engineering, Science and Health Systems		
PI Address 2:			
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Astronauts experience decreased cellular immunity during spaceflight. To successfully achieve long-term manned spaceflight, countermeasures must be developed to ameliorate immune function during these missions. The design of effective countermeasures will require a detailed knowledge of the underlying molecular mechanisms responsible for decreased immunity in spaceflight. T lymphocytes are the principle mediators of cellular immunity; T lymphocytes are insensitive to mitogenic stimulation when cultured in vitro during orbital flight. Inhibited activation of these cells can be overcome by using an in-flight centrifuge to produce artificial 1-g conditions; a partial recovery of activation can be achieved by direct activation of second-messenger signaling via phorbol ester and calcium ionophore. These results indicate a gravity dependent defect in signal transduction upstream of the phosphatidyl inositol (PIP2) second messenger pathway. This project investigates the hypothesis that microgravity prevents T cell activation by interfering with membrane-proximal signal transduction through the T cell receptor (TCR). Specifically, this study examines the affect		

	of microgravity on the mechanisms that lead to the activation of phospholipase C-g1 (PLC-g1) and subsequent activation of PIP2 signaling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In vitro culture of T cells in true- and modeled-microgravity provides a unique (non-chemical, epigenetic) means of inhibiting signal transduction; allowing for a minimally invasive examination of the key signaling molecules required for T cell activation. Identification of these molecules in fully functional cells in a severely inhibitory environment will also provide insight into T cell function in less inhibitory situations such as aging. Conversely, identification of key signaling molecules could also provide novel targets for treatment of autoimmune disease.
Task Progress:	The goal of the project described here is to investigate the impairment of in vitro T cell activation observed during spaceflight using culture of T lymphocytes in rotating wall vessel (RWV) bioreactors as a ground-based model of microgravity (MMG). The working hypothesis behind this research is that T lymphocytes fail to become activated in true- and modeled-microgravity due to a disruption of membrane-proximal signal transduction. Our goal in testing this hypothesis is to identify the underlying molecular mechanisms responsible for the inhibitory affect of MMG on T lymphocyte activation. The first aim of this study was to examine the kinetics of MMG-induced inhibition of T cell activation. In this regard we found that murine T lymphocytes stimulated in the RWV are affected solely by dysfunctional signal transduction only during the first 8-hrs of culture. Stimulation in the RWV for longer durations resulted in the activation of secondary mechanisms that resulted in cell death. These mechanisms were found to be due to the RWV culture environment rather that due to the prolonged lack of an activating signal since cells left unstimulated at 1-g for similar durations did not experience the same loss of viability. The ability of the cells to transduce activating signals appeared to remain intact following all durations of culture in the RWV, as even the few remaining viable cells in cultures exposed to extended durations of MMG were fully capable of upregulating the IL-2R a-chain when returned to 1-g conditions. This finding is comparable to similar studies examining the ability of human T lymphocytes to become activated following culture in the RWV, however, the loss of viability appears to be species specific as human T cells can endure culture in the RWV for up to 72-hrs without a significant loss of viable cells. The final aim of this project is to examine TCR signal transduction in the RWV at the molecular level in order to determine the gravity sensitive mechanisms within this pathway. Signal transduction through the TCR proceeds via a tyrosine kinase signaling pathway. Typically, receptor tyrosine kinase pathways are initiated by intrinsic kinase activity within the intracellular portion of the receptor itself. The TCR has no intrinsic kinase activity; instead, upon TCR engagement the src-family kinase lck initiates signal transduction by phosphorylating several sites within the intracellular portions of the TCR/CD3 complex. The signal is then propagated when a second kinase, ZAP-70, binds these sites and is activated via phosphorylation by lck. ZAP-70 has several substrates including LAT, a non-enzymatic adaptor molecule that serves as the nucleation point for transmission of the signal away from the membrane—primarily via recruitment of PLC-g1 and PI3K. Based on our preliminary data and reports in the literature, T cells stimulated in the RWV are unable to transmit signals away from the membrane by PLC-g1 mediated mechanisms. To investigate the initiation of these events by TCR stimulation in the RWV we have recently optimized a protocol to monitor the activation of ZAP-70 by phosphorylation of tyrosine 493 using flow cytometry. Under static conditions this residue is rapidly phosphorylated by lck, and is required for the enzymatic activity of ZAP-70. We have found that in a transformed T cell line (Jurkat) ZAP-70 becomes phosphorylated at this site in response to TCR stimulation in the RWV. We are currently working to expand this analysis to monitor the activation status of LAT and PLC-g1 following stimulation in the RWV, and also to perform identical studies using primary human T cells. Jurkat T cells offer an interesting basis of comparison since they are hypersensitive to TCR stimulation due to defects in phospholipid metabolism in the membrane. If, as preliminary results suggest, Jurkat T cells are capable of signal transduction in the RWV whereas primary T cells are not, the phospholipid content of the membrane may provide a convenient point for intervention in the search for a means of overcoming defects in TCR signal transduction during microgravity culture. Jurkat T cells lack the lipid phosphatase PTEN (phophatase and tenson homologue deleted on chromosome ten), which leads to an accumulation of phosphatidyl inositol 3,4,5-trisphosphate (PIP3)in the plasma membrane. This phospholipid is normally only present in activated cells, and its presence in the membrane recruits several enzymes to the membrane in response to TCR engagement. PLC-g1 can be recruited to the membrane and tyrosine phosphorylation in the absence of PIP3, however, enzymatic activity requires the presence of PIP3.
Bibliography Type:	Description: (Last Updated: 01/15/2008)
Articles in Peer-reviewed Journals	Simons DM, Gardner EM, Lelkes PI. "Dynamic culture in a rotating-wall vessel bioreactor differentially inhibits murine T-lymphocyte activation by mitogenic stimuli upon return to static conditions in a time-dependent manner." J Appl Physiol. 2006 Apr;100(4):1287-92. PMID: 16384837 , Apr-2006